Programming for Problem Solving Using Python
Unit-1

Session 1: Local Environment Setup, Setting up Python environment: Installing Python, IDEs (Installation Guide)

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Session-1

Session-1: Local Environment Setup, Setting up Python environment: Installing Python, IDEs

Introduction to Python

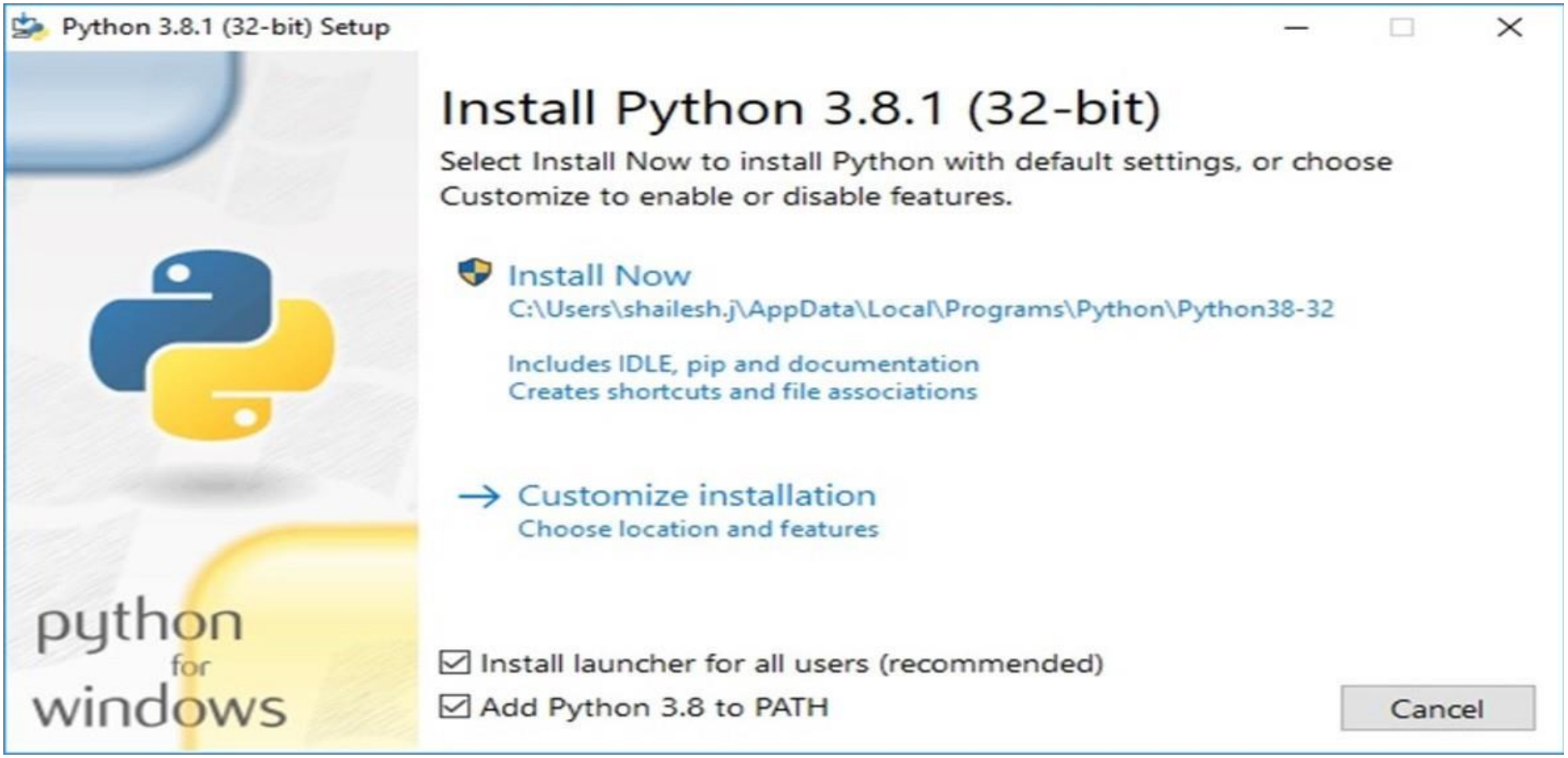
- Python is a **high-level, interpreted** programming language.
- Developed by **Guido van Rossum**, first released in **1991**.
- Known for its **readability**, simplicity, and versatile syntax.

- Widely used in:
  - **Web Development** (e.g., Django, Flask)
  - **Data Science & Machine Learning** (e.g., Pandas, NumPy, scikit-learn, TensorFlow)
  - **Automation/Scripting** (e.g., file handling, task automation)
  - **Software Development, APIs, DevOps, Cybersecurity**, and more.

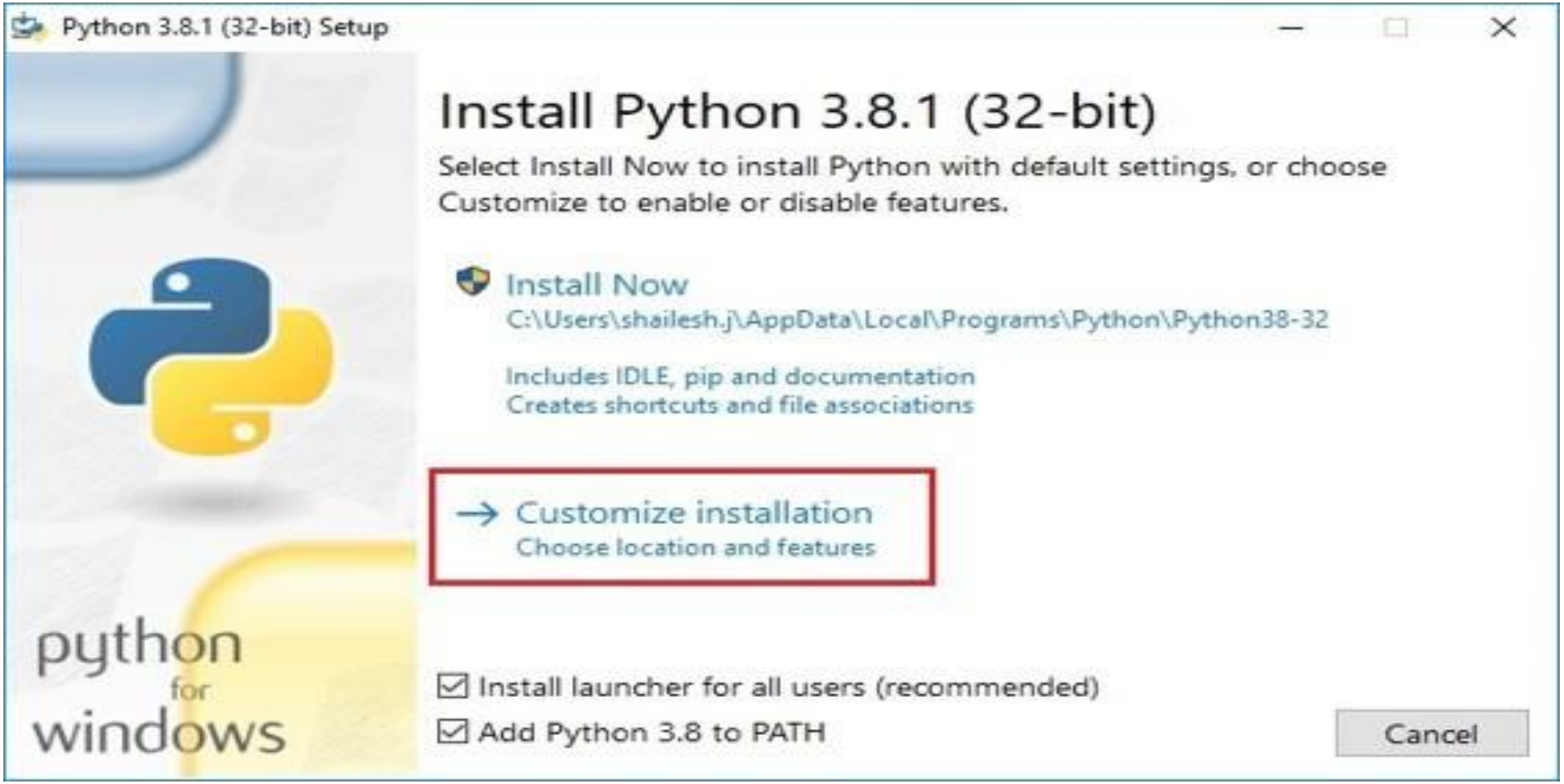
Session-1: Installing Python on Windows

- Visit the link https://www.python.org/downloads/ to download the latest release of Python.
- Latest version:3.8.5
- https://www.python.org/downloads/windows/ Download
- Windows x86-64 executable installer

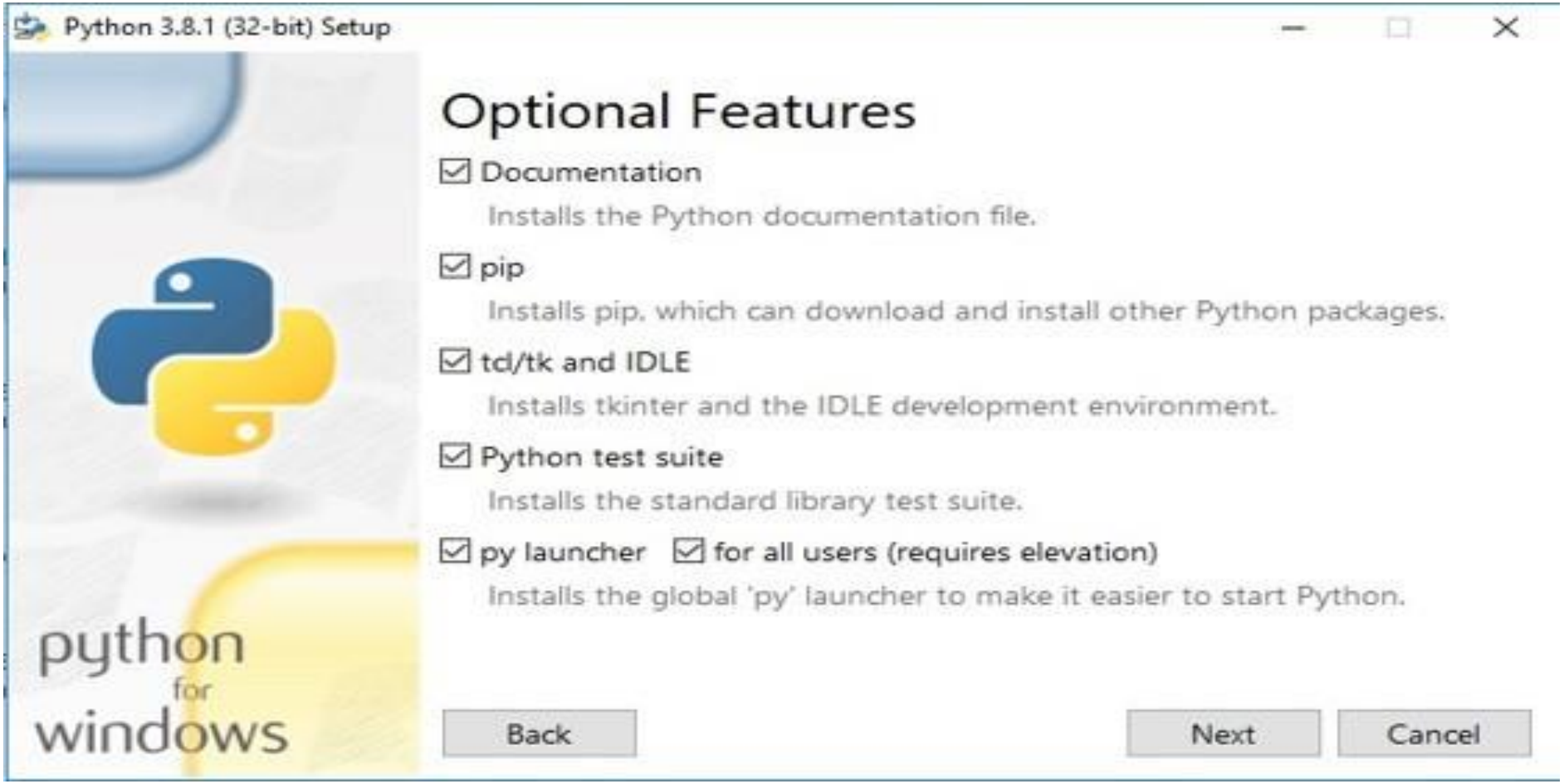
Session-1: Installing Python on Windows



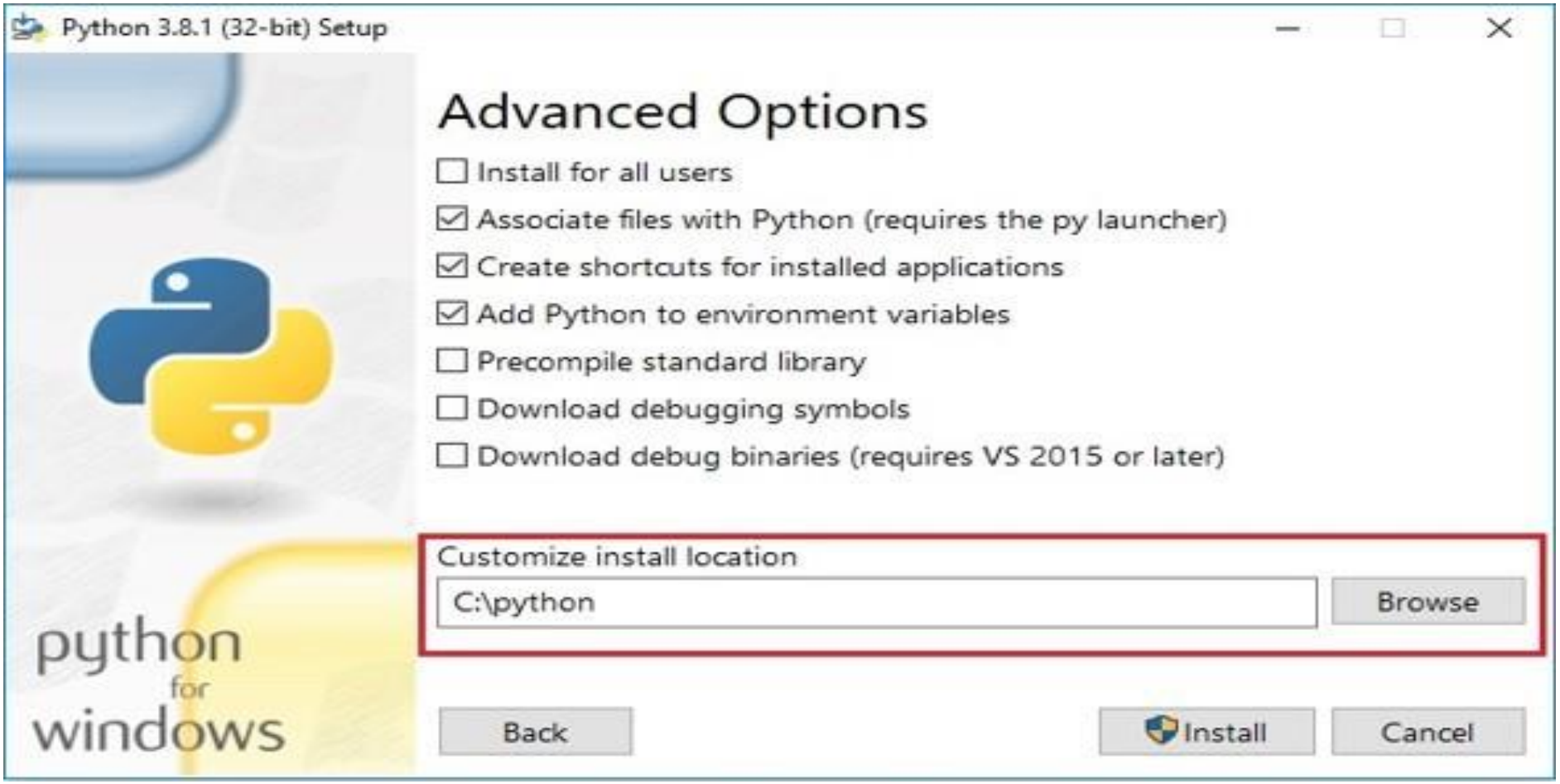
Session-1: Installing Python on Windows



Session-1: Installing Python on Windows



Session-1: Installing Python on Windows



– Check the version of Python installer

python –version

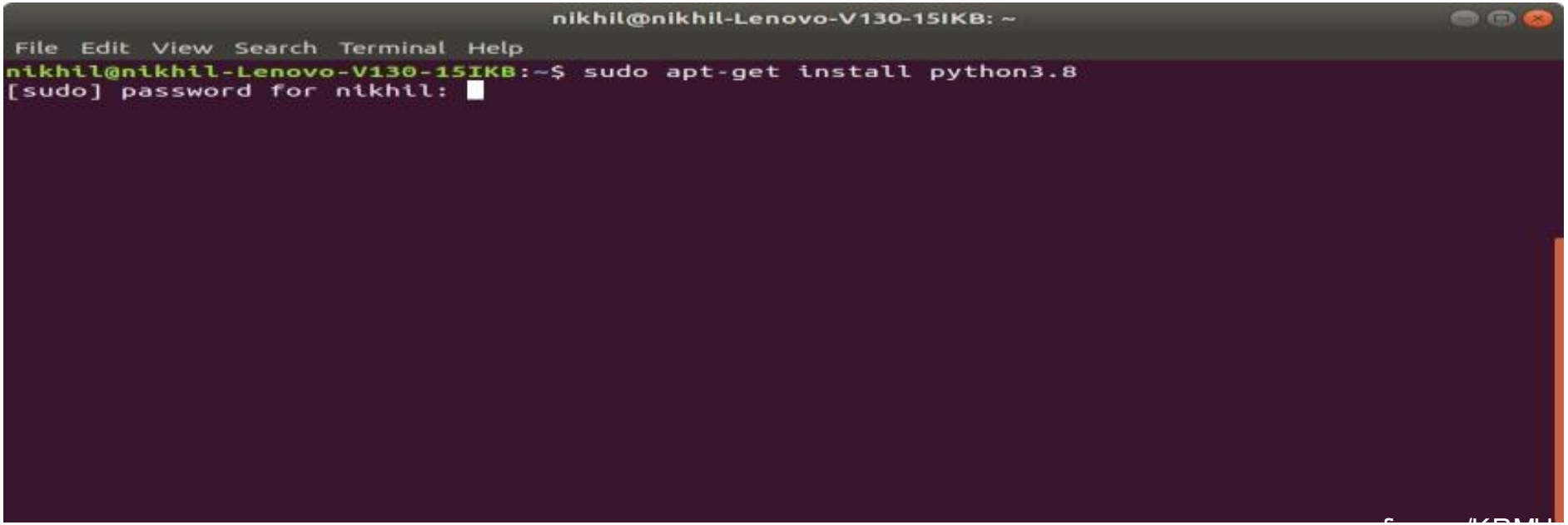
Session-1: Installing Python on Windows

– To check the version of Python installed, use this command or cmd.

Session-1: Installing Python on Linux

Beginning Python installation
For almost every Linux system, the following command could be used to install Python directly:

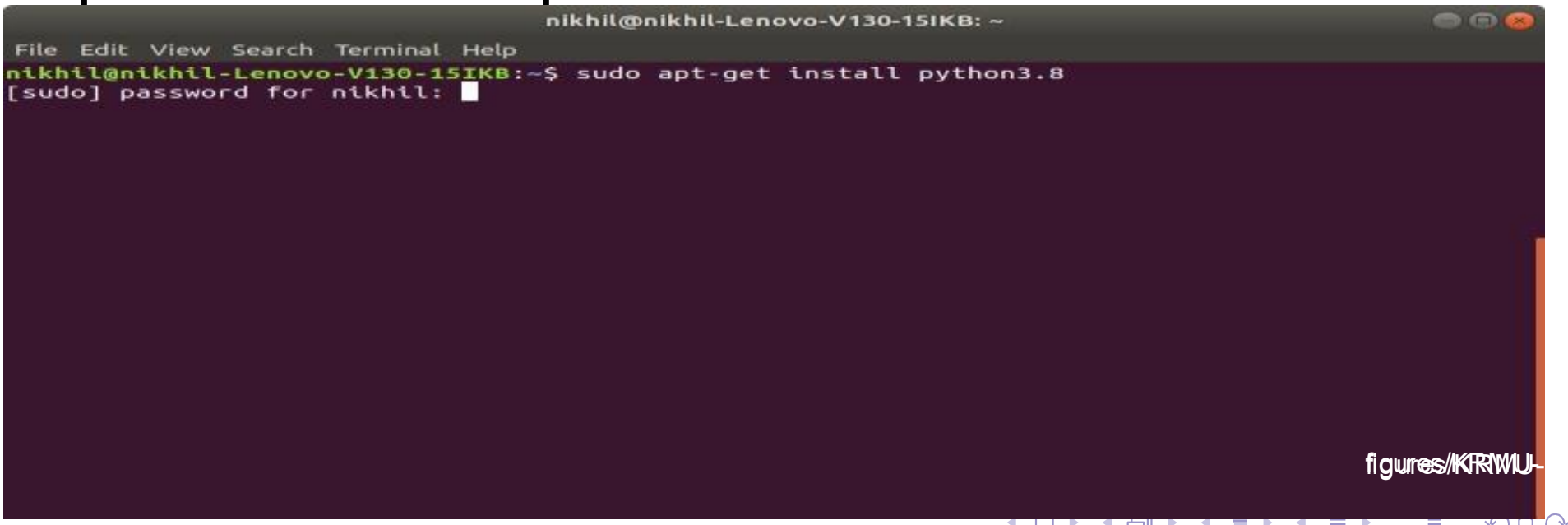
$sudo apt-get install python3.8



Session-1: Installing Python on Linux (contd.)

Getting Started

We will start our installation by writing the password as mentioned in the below picture that gives the permission to perform further operations.



figures/KRIWU-

Session-1: Installing Python on Linux (contd..)

Assigning DiskSpace
Generally, it asks for additional diskspace and we generally continue by typing Y and then enter.

figures/KRIWU-

Session-1: Installing Python on Linux (contd...)

Finished Installation

Here, the required installation is completed.

```
nikhil@nikhil-Lenovo-V130-15IKB: ~
File  Edit  View  Search  Terminal  Help
-3~18.04 [355 kB]
Fetched 4,551 kB in 3s (1,427 kB/s)
Selecting previously unselected package libpython3.8-minimal:amd64.
(Reading database ... 258857 files and directories currently installed.)
Preparing to unpack .../libpython3.8-minimal_3.8.0-3~18.04_amd64.deb ...
Unpacking libpython3.8-minimal:amd64 (3.8.0-3~18.04) ...
Selecting previously unselected package python3.8-minimal.
Preparing to unpack .../python3.8-minimal_3.8.0-3~18.04_amd64.deb ...
Unpacking python3.8-minimal (3.8.0-3~18.04) ...
Selecting previously unselected package libpython3.8-stdlib:amd64.
Preparing to unpack .../libpython3.8-stdlib_3.8.0-3~18.04_amd64.deb ...
Unpacking libpython3.8-stdlib:amd64 (3.8.0-3~18.04) ...
Selecting previously unselected package python3.8.
Preparing to unpack .../python3.8_3.8.0-3~18.04_amd64.deb ...
Unpacking python3.8 (3.8.0-3~18.04) ...
Setting up libpython3.8-minimal:amd64 (3.8.0-3~18.04) ...
Setting up python3.8-minimal (3.8.0-3~18.04) ...
Setting up libpython3.8-stdlib:amd64 (3.8.0-3~18.04) ...
Setting up python3.8 (3.8.0-3~18.04) ...
Processing triggers for gnome-menus (3.13.3-11ubuntu1.1) ...
Processing triggers for mime-support (3.60ubuntu1) ...
Processing triggers for desktop-file-utils (0.23-1ubuntu3.18.04.2) ...
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
nikhil@nikhil-Lenovo-V130-15IKB:~$
```

Session-1: Installing Python on Linux (contd....)

To verify the installation enter the following commands in your Terminal.

$python3.8

Session-1: Installing Python on MacOS

Visit the Python Website.
Go to the official Python website https://www.python.org/downloads/macos/ to access the download page for the latest version of Python for macOS.

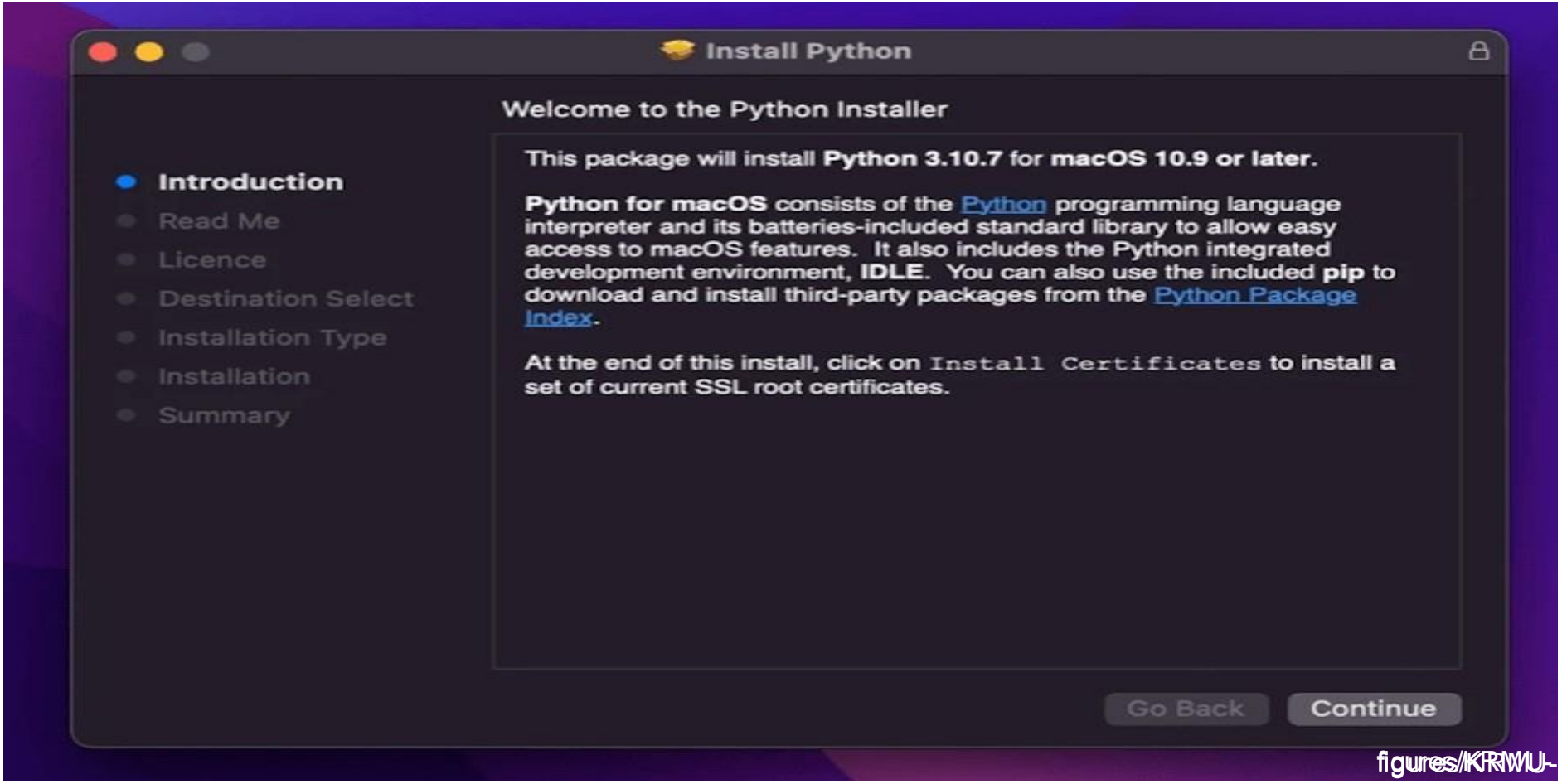
Session-1: Installing Python on MacOS (contd.)

Download the macOS Installer.
-> On the download page, you'll find the macOS installer package (.pkg file) for the most recent Python release. Download the installer to your computer.

Session-1: Installing Python on MacOS (contd..)

–> Run the Installer and Follow the Instructions.
–> Locate the downloaded installer file (usually in your Downloads folder) and double-click on it to run the installation process.
–> Proceed through the installation steps by agreeing to the software license agreement, choosing the installation location (we recommend using the default location), and entering your administrator password when prompted.

Session-1: Installing Python on MacOS (contd...)



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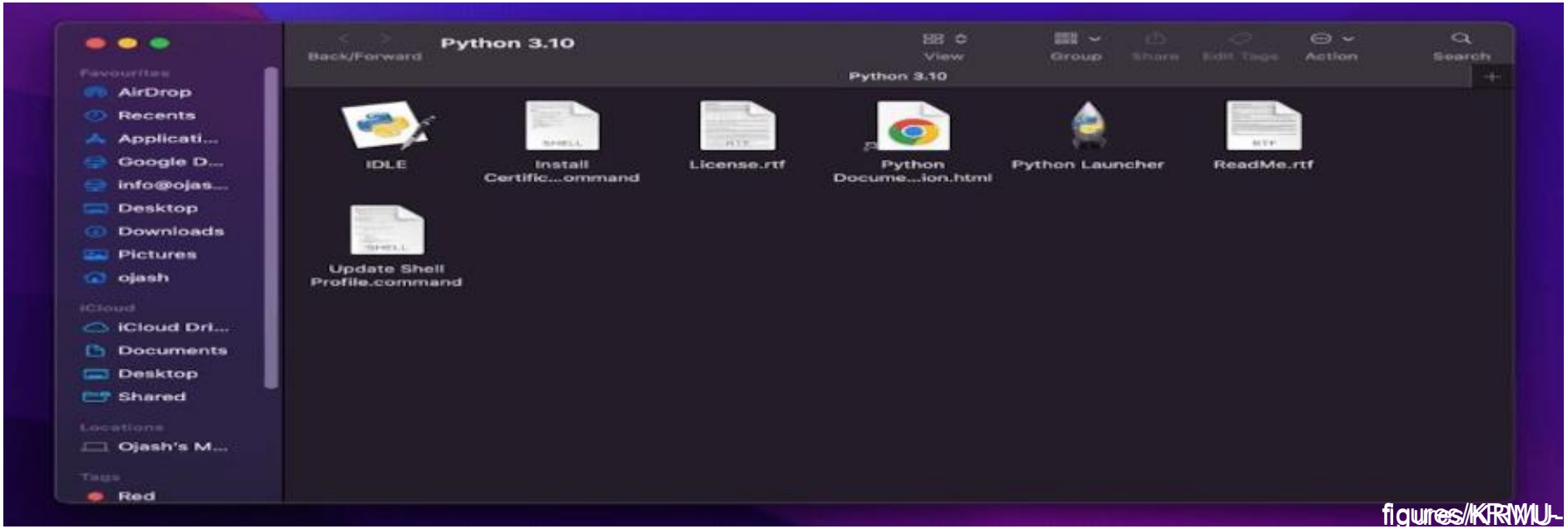
Session-1: Installing Python on MacOS (contd....)

Verify Python and IDLE Are Installed Correctly.
–> Once the installation process is completed, a folder will open on your desktop. Click IDLE in this folder.



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Session-1: Installing Python on MacOS (contd....)

IDLE is the independent development environment that comes with Python.
Once opened, you should see a Python shell automatically.

```
Python 3.10.7 (v3.10.7:6cc6b13308, Sep  5 2022, 14:02:52) [Clang 13.0.0 (clang-1
300.0.29.30)] on darwin
Type "help", "copyright", "credits" or "license()" for more information.
|
```

Session-1: Installing Python on MacOS (contd....)

Verify the Installation with Terminal
You can verify installation via the Terminal as well. Open the Terminal app and type the following command:

```
python3 --version
```

Press Enter, and you should see the version of Python you just installed displayed.
This confirms that Python has been successfully installed on your Mac.

Session-1 : Test Your Knowledge- Fill in the blanks

- Python 2 reached its end-of-life in _____, and it is no longer maintained.
- _____is an open-source Python IDE designed specifically for scientific computing and data analysis.
- _____is a lightweight and highly customizable code editor that supports Python development with various plugins.
- _____is Python's built-in IDE, which comes bundled with the standard Python distribution.

Session-1: Test Your Knowledge- MCQ

- What feature is NOT typically included in an IDE?
  1. Source code editor
  2. Debugger
  3. Build automation tools
  4. Version control system
- What is the main advantage of using virtual environments with Python IDEs?
  1. Improved code completion
  2. Better integration with version control systems
  3. Dependency isolation
  4. Enhanced debugging capabilities

Session-1: Test Your Knowledge- MCQ (Solutions)

- What feature is NOT typically included in an IDE?

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- What is the main advantage of using virtual environments with Python IDEs?

  1. Improved code completion
  2. Better integration with version control systems
  3. Dependency isolation
  4. Enhanced debugging capabilities

Session-1: Test Your Knowledge- Interview Questions (Solutions)

1. Python 3.11.3.
2. IDLE stands for Integrated Development and Learn- ing Environment. It is a simple, lightweight environ- ment provided with the default Python installation, which includes an editor and a Python shell for exe- cuting code.
3. When you see >>> inside IDLE, it indicates that you are at the Python interactive shell prompt.

Session-1 : Test Your Knowledge- Interview Questions (Solutions)

4. Python uses indentation to define the structure of the code, such as the bodies of loops, functions, and conditionals.
5. python –version

Session-1: Test Your Knowledge- Interview Questions (Solutions)

6. PyCharm, Visual Studio Code (VS Code), Jupyter Notebook, Spyder, IDLE
7. Libraries: NumPy, Pandas, Matplotlib, Requests
   Packages: SciPy, Flask, Django, Scikit-Learn

Thank You!

Any Questions?